THE LARGEST KEMENY CONSTANT AND THE LARGEST RELAXATION TIME OF A GRAPH

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Abstract. For the simple random walk on a connected graph with n vertices, let τ_0 be Kemeny's constant and τ_2 the relaxation time. Aldous and Fill conjectured that the maxima of both quantities over all connected n-vertex graphs are asymptotic to $n^3/54$. We prove this. Every connected graph on n vertices satisfies $\tau_2 \le \tau_0 \le n^3/54 + \frac{37}{8}n^{5/2}$, and the barbell made of two cliques on about $n/3$ vertices joined by a path on about $n/3$ vertices has $\tau_2 \ge (n-2)(n-5)(n+3)/54$. The upper bound comes from the electrical formula for Kemeny's constant: the vertices near high-degree vertices are grouped into components of small total diameter, the rest of the graph is compared with a tree, and each edge of the tree is shown to contribute at most $s^2/8$ plus a lower-order term, where s is the number of grouped vertices.

1. Introduction

Let $G = (V, E)$ be a finite connected simple graph with $n \ge 2$ vertices, degrees d_v, and total degree $W = \sum_v d_v = 2|E|$. The simple random walk on G has transition matrix $P(u, v) = 1/d_u$ for $uv \in E$ and stationary distribution $\pi_v = d_v/W$. We run it in continuous time with generator $P - I$, so that it jumps at rate one; expected hitting times are the same as for the discrete-time walk. Let T_v be the hitting time of v, with $T_v = 0$ for a walk started at v, and let

$$0 = \mu_1 < \mu_2 \le \cdots \le \mu_n$$

be the eigenvalues of $I - P$. Following Aldous and Fill [1], put

$$\tau_0 = \sum_{v \in V} \pi_v E_u[T_v], \qquad \tau_2 = \frac{1}{\mu_2}.$$

The sum τ_0 does not depend on u; it is Kemeny's constant of the walk [7]. The relaxation time τ_2 is the inverse spectral gap. Some authors define Kemeny's constant with an extra term 1, or for the lazy walk $(I + P)/2$, which doubles both τ_0 and τ_2; the constant $1/54$ below refers to the normalisation just given.

Among the extremal questions collected by Aldous and Fill, the largest hitting time and the largest cover time of an n-vertex graph are of order n^3, with constant $4/27$, attained asymptotically by lollipop graphs (Brightwell and Winkler [3]; Feige [6]). For τ_0 and τ_2 the natural candidates are barbells: two cliques joined by a long path. Aldous and Fill [1, Open Problem 6.13] conjecture that

$$\max_G \tau_0(G) \sim \frac{n^3}{54} \quad \text{and} \quad \max_G \tau_2(G) \sim \frac{n^3}{54},$$

the maxima being over connected graphs with n vertices. Breen, Butler, Day, DeArmond, Lorenzen, Qian and Riesen [2] computed Kemeny's constant of barbell-type graphs and showed that it is of order n^3, the largest possible order. Faught, Kempton and Knudson [5] obtained

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Kemeny's constant of a barbell from a formula for graphs with a cut vertex and showed that the best barbell has cliques on about $n/3$ vertices each and Kemeny's constant $n^3/54 + O(n^2)$; they note that barbells are believed, on the basis of computation, to maximise Kemeny's constant. The matching upper bound is the content of the present paper.

Theorem 1.1. *Let G be a connected simple graph on $n \geq 2$ vertices. Then*

$$\tau_2(G) \leq \tau_0(G) \leq \frac{n^3}{54} + \frac{37}{8}\, n^{5/2}. \tag{1.1}$$

For $n \geq 6$ let $a = \lfloor n/3 \rfloor$ and $\ell = n - 2a + 1$, and let B_n be the barbell consisting of two disjoint copies of K_a whose chosen vertices are joined by a path with ℓ edges. Then B_n has n vertices and

$$\tau_0(B_n) \geq \tau_2(B_n) \geq \frac{3a(a-1)\ell + \ell^2 + 2}{6} \geq \frac{(n-2)(n-5)(n+3)}{54}. \tag{1.2}$$

Corollary 1.2. *As $n \to \infty$,*

$$\max_G \tau_0(G) = \frac{n^3}{54} + O(n^{5/2}), \qquad \max_G \tau_2(G) = \frac{n^3}{54} + O(n^{5/2}),$$

the maxima being over connected simple graphs on n vertices, and the barbells B_n attain the leading constant for both quantities.

The constant has a simple explanation, and the proof follows it. In a barbell with cliques of size a and a path of ℓ edges, each path edge separates two halves of total degree about a^2 each, and the electrical formula (2.3) below charges each such edge about $a^2 \cdot a^2/(2a^2) = a^2/2$. The total, $a^2\ell/2$ with $2a + \ell \approx n$, is largest at $a = \ell = n/3$, where it equals $n^3/54$. For a general graph we single out the set S of vertices within distance one of a vertex of degree at least $\sqrt{n}$. Its components have small total diameter, every other vertex has small degree, and after the components are contracted the graph is compared with a tree. A tree edge separates the $s = |S|$ grouped vertices into two parts of sizes x and y, and its contribution is at most $x^2y^2/(x^2+y^2) \leq s^2/8$, up to a term of order $n^{3/2}$. There are at most $n - s$ such edges, up to lower-order terms, and $(n-s)s^2/8 \leq n^3/54$.

The theorem leaves open the exact maximiser for each n and the true size of the error term. The best barbell has $\tau_0 = n^3/54 + O(n^2)$ by [5], which suggests that the error term in (1.1) can be reduced to $O(n^2)$.

Section 2 recalls the spectral and electrical formulas for τ_0. Section 3 proves the upper bound (1.1), Section 4 the lower bound (1.2), and Section 5 reports a numerical check.

2. Spectral and electrical formulas

Let $L = \operatorname{diag}(d) - A$ be the combinatorial Laplacian of G, where A is the adjacency matrix. The matrix $I - P = \operatorname{diag}(d)^{-1}L$ is similar to the symmetric matrix $\operatorname{diag}(d)^{-1/2} L \operatorname{diag}(d)^{-1/2}$, and since $f^{\mathsf{T}} L f = \sum_{uv \in E} (f_u - f_v)^2$ and G is connected, its eigenvalue 0 is simple. The variational characterisation gives

$$\mu_2 = \min\Big\{ \frac{\sum_{uv \in E} (f_u - f_v)^2}{\sum_v d_v f_v^2} : \; f \neq 0, \; \sum_v d_v f_v = 0 \Big\}. \tag{2.1}$$

Lemma 2.1. *For the simple random walk on a connected graph,*

$$\tau_0 = \sum_{i=2}^{n} \frac{1}{\mu_i}, \tag{2.2}$$

and in particular $\tau_2 \le \tau_0$. *Moreover*

$$\tau_0 = \frac{1}{W} \sum_{\{u,v\} \subset V} d_u d_v R_{uv}, \tag{2.3}$$

where the sum is over unordered pairs of distinct vertices and R_{uv} *is the effective resistance between* u *and* v *when every edge has unit resistance. Finally* $R_{uv} \le \operatorname{dist}_G(u,v)$.

Proof. Identity (2.2) is the eigentime identity [1, Chapter 3]; it also follows from the fundamental matrix $Z = (I - P + \mathbf{1}\pi^{\mathsf{T}})^{-1}$, which has eigenvalues 1 and $1/\mu_i$ ($i \ge 2$) and satisfies $E_u[T_v] = (Z_{vv} - Z_{uv})/\pi_v$ [7], so that $\tau_0 = \operatorname{tr} Z - \sum_v Z_{uv} = \operatorname{tr} Z - 1$. Since every term of (2.2) is positive, $\tau_2 = 1/\mu_2 \le \tau_0$.

For (2.3), average the definition of τ_0 over the starting vertex u with respect to π:

$$\tau_0 = \sum_{u,v} \pi_u \pi_v E_u[T_v] = \frac{1}{2} \sum_{u,v} \pi_u \pi_v (E_u[T_v] + E_v[T_u]).$$

By the commute time identity [4], $E_u[T_v] + E_v[T_u] = W R_{uv}$. Hence $\tau_0 = \frac{W}{2} \sum_{u,v} \pi_u \pi_v R_{uv} = \frac{1}{2W} \sum_{u,v} d_u d_v R_{uv}$, which is (2.3).

By Thomson's principle, R_{uv} is the minimum energy $\sum_e j_e^2$ of a unit flow j from u to v. The unit flow along a shortest path from u to v has energy $\operatorname{dist}_G(u,v)$, so $R_{uv} \le \operatorname{dist}_G(u,v)$. □

3. The upper bound

Throughout this section G is a connected simple graph on n vertices and $D > 0$ is a threshold. Let

$$H = \{v \in V : d_v \ge D\}, \qquad S = \bigcup_{h \in H} N[h],$$

where $N[h]$ is the closed neighbourhood of h. Let $C_1, \dots, C_k$ be the vertex sets of the components of the induced graph $G[S]$, let $s_i = |C_i|$, $s = |S| = \sum_i s_i$, and let δ_i be the diameter of the induced graph $G[C_i]$. Put $\delta = \sum_i \delta_i$. If H is empty, then S is empty and $s = k = \delta = 0$.

Lemma 3.1. *We have* $k \le s/D$ *and* $\delta \le 5s/D$.

Proof. Each C_i contains a vertex $h \in H$ together with all of $N[h]$, because $N[h] \subset S$ is connected. Hence $s_i \ge |N[h]| > D$ and $k \le s/D$.

Fix i and a shortest path $v_0, v_1, \dots, v_{\delta_i}$ in $G[C_i]$. For each index $j \in \{0, 5, 10, \dots\}$ with $j \le \delta_i$, choose $h_j \in H$ with $v_j \in N[h_j]$; such a vertex exists because $v_j \in S$, and then $N[h_j] \subset C_i$. If $N[h_j]$ and $N[h_m]$ met for two distinct selected indices $j < m$, the walk from v_j through h_j, a common vertex of the two neighbourhoods and h_m to v_m would have length at most four and would stay in C_i, while $\operatorname{dist}_{G[C_i]}(v_j, v_m) = m - j \ge 5$. So the sets $N[h_j]$ are pairwise disjoint subsets of C_i, each with more than D elements, and

$$\Big(\Big\lfloor \frac{\delta_i}{5} \Big\rfloor + 1 \Big) D \le s_i.$$

Since $\delta_i/5 < \lfloor \delta_i/5 \rfloor + 1$, this gives $\delta_i \le 5 s_i / D$. Summing over i gives $\delta \le 5s/D$. □

Contract each C_i to a single vertex and keep every vertex outside S. The resulting graph Q is connected and has $k + (n - s)$ vertices; let $q : V \to V(Q)$ be the quotient map. Fix a spanning tree T of Q. It has

$$|E(T)| = k + n - s - 1 \le n - s + \frac{s}{D} \tag{3.1}$$

edges, and also $|E(T)| \le n - 1$. For a vertex x of Q let $w_x = \sum_{v \in q^{-1}(x)} d_v$, so that $\sum_x w_x = W$. For an edge e of T, let $W_L(e)$ and $W_R(e)$ be the sums of w_x over the two components of $T - e$.

Lemma 3.2. *We have*

$$\tau_0 \le \frac{1}{W}\sum_{e\in E(T)} W_L(e)W_R(e) + \frac{W\delta}{2}.$$

Proof. Let $u, v \in V$ and let $x_0 = q(u), x_1, \dots, x_m = q(v)$ be the path from $q(u)$ to $q(v)$ in T. Each edge $x_{r-1}x_r$ of T comes from an edge of G between the parts $q^{-1}(x_{r-1})$ and $q^{-1}(x_r)$. Joining these m edges by paths inside the contracted components, and adding a path inside the component of u (or of v) when u (or v) lies in S, gives a walk from u to v in G. The tree path visits each contracted component at most once, and inside C_i the walk uses a path of length at most δ_i. Therefore

$$\operatorname{dist}_G(u,v) \le \operatorname{dist}_T(q(u),q(v)) + \delta.$$

By Lemma 2.1,

$$\tau_0 \le \frac{1}{W}\sum_{\{u,v\}} d_u d_v \operatorname{dist}_T(q(u),q(v)) + \frac{\delta}{W}\sum_{\{u,v\}} d_u d_v.$$

The second term is at most $\delta W^2/(2W) = W\delta/2$. In the first, pairs inside one part contribute nothing, and grouping the rest by parts gives $\sum_{\{x,y\}} w_x w_y \operatorname{dist}_T(x,y)$. An edge e of T lies on the tree path between x and y exactly when it separates them, so this sum equals $\sum_e W_L(e)W_R(e)$. □

Lemma 3.3. *Every edge e of T satisfies*

$$\frac{W_L(e)W_R(e)}{W} \le \frac{s^2}{8} + 2nD.$$

Proof. For the vertex of Q obtained from C_i put $a_x = 2|E(G[C_i])|$, and for a vertex $x \notin S$ put $a_x = 0$. Let $b_x = w_x - a_x \ge 0$, $A = \sum_x a_x$ and $B = \sum_x b_x$, so that $W = A + B$. Two different components C_i, C_j are not joined by an edge of G, so every edge of G that does not lie inside one of the sets C_i has an endpoint outside S. The number B counts the endpoints of such edges, and therefore

$$B \le 2\sum_{v\notin S} d_v < 2nD,$$

because every vertex outside S is outside H.

Fix e and write A_L, b_L and A_R, b_R for the sums of a_x and b_x on the two sides, so that $W_L = A_L + b_L$ and $W_R = A_R + b_R$. Then

$$W_L W_R = A_L A_R + (A_L b_R + A_R b_L + b_L b_R) \le A_L A_R + AB + B^2 = A_L A_R + BW.$$

If $A = 0$ then $A_L A_R = 0$ and the claim follows. Otherwise $W \ge A$ gives

$$\frac{W_L W_R}{W} \le \frac{A_L A_R}{A_L + A_R} + B.$$

Let x and y be the numbers of vertices of S on the two sides of e, so $x + y = s$. A simple graph on s_i vertices has fewer than s_i^2 edge endpoints, hence $A_L \le \sum s_i^2 \le x^2$, the sum being over the components on the left, and likewise $A_R \le y^2$. The function $(p,q) \mapsto pq/(p+q)$ is increasing in each variable on $p, q \ge 0$ (read as 0 when $p + q = 0$), so

$$\frac{A_L A_R}{A_L + A_R} \le \frac{x^2 y^2}{x^2 + y^2} \le \frac{s^2}{8},$$

where the last inequality is $(x+y)^2(x^2+y^2) - 8x^2y^2 = (x-y)^2(x^2+4xy+y^2) \ge 0$. □

Proof of the upper bound (1.1). The inequality $\tau_2 \le \tau_0$ is in Lemma 2.1. By Lemmas 3.2, 3.3 and 3.1, together with (3.1) and $|E(T)| \le n - 1$,

$$\tau_0 \le \Big(n - s + \frac{s}{D}\Big)\frac{s^2}{8} + (n-1)\cdot 2nD + \frac{5Ws}{2D}.$$

Since $s \le n$ and $W \le n^2$, the right side is at most

$$\frac{(n-s)s^2}{8} + \frac{n^3}{8D} + 2n^2D + \frac{5n^3}{2D} = \frac{(n-s)s^2}{8} + \frac{21}{8}\cdot\frac{n^3}{D} + 2n^2D.$$

On $0 \le s \le n$ the function $(n-s)s^2$ has its maximum $4n^3/27$ at $s = 2n/3$, so $(n-s)s^2/8 \le n^3/54$. Taking $D = \sqrt{n}$ gives

$$\tau_0 \le \frac{n^3}{54} + \frac{21}{8}n^{5/2} + 2n^{5/2} = \frac{n^3}{54} + \frac{37}{8}n^{5/2}. \qquad \square$$

Remark 3.4. The maximum of $(n-s)s^2$ at $s = 2n/3$ matches the barbell: two cliques on $n/3$ vertices each make up the set S, and the path, with about $n/3$ edges, is the tree outside it. The proof bounds every tree edge as if it were the middle edge of the path; for the barbell this costs nothing, because every path edge has the two cliques on opposite sides.

4. The lower bound

Let $a \ge 2$ and $\ell \ge 1$, and let $B(a,\ell)$ be the graph formed by two disjoint copies of K_a and a path with ℓ edges joining a vertex p_0 of the first copy to a vertex p_ℓ of the second, through $\ell - 1$ new vertices $p_1, \dots, p_{\ell-1}$. It has $2a + \ell - 1$ vertices.

Proposition 4.1. $\tau_2(B(a,\ell)) \ge \dfrac{3a(a-1)\ell + \ell^2 + 2}{6}$.

Proof. Define $f = -1$ on the first clique, $f = 1$ on the second, and $f(p_j) = -1 + 2j/\ell$ for $0 \le j \le \ell$; the two definitions agree at p_0 and p_ℓ. The reflection of $B(a,\ell)$ that exchanges the cliques and maps p_j to $p_{\ell-j}$ preserves degrees and changes the sign of f, so $\sum_v d_v f_v = 0$, and (2.1) gives

$$\tau_2 = \frac{1}{\mu_2} \ge \frac{\sum_v d_v f_v^2}{\sum_{uv\in E}(f_u - f_v)^2}.$$

Only the ℓ path edges contribute to the denominator, each with $(2/\ell)^2$, so the denominator equals $4/\ell$. In the numerator, each clique has total degree $a(a-1)+1$ and $f^2 = 1$ on it, and the $\ell - 1$ inner path vertices have degree 2, so the numerator equals

$$2a(a-1) + 2 + 2\sum_{j=1}^{\ell-1}\Big(1 - \frac{2j}{\ell}\Big)^2 = 2a(a-1) + 2 + \frac{2(\ell-1)(\ell-2)}{3\ell} = 2a(a-1) + \frac{2\ell}{3} + \frac{4}{3\ell},$$

using $\sum_{j=1}^{\ell-1}(1 - 2j/\ell)^2 = (\ell-1) - 2(\ell-1) + \frac{2(\ell-1)(2\ell-1)}{3\ell} = \frac{(\ell-1)(\ell-2)}{3\ell}$. Multiplying by $\ell/4$ gives the claim. $\square$

Proof of the lower bound (1.2). For $n \ge 6$, $a = \lfloor n/3 \rfloor \ge 2$ and $\ell = n - 2a + 1 \ge n/3 + 1$, and $B_n = B(a,\ell)$ has $2a + \ell - 1 = n$ vertices. The first inequality in (1.2) is Lemma 2.1 and the second is Proposition 4.1. For the third, $a \ge (n-2)/3$ gives $a(a-1) \ge (n-2)(n-5)/9$, and $\ell \ge (n+3)/3$, so

$$\frac{3a(a-1)\ell + \ell^2 + 2}{6} \ge \frac{a(a-1)\ell}{2} \ge \frac{(n-2)(n-5)(n+3)}{54}. \qquad \square$$

Proof of Corollary 1.2. For $n \ge 6$, Theorem 1.1 gives

$$\frac{(n-2)(n-5)(n+3)}{54} \le \tau_2(B_n) \le \max_G \tau_2(G) \le \max_G \tau_0(G) \le \frac{n^3}{54} + \frac{37}{8}n^{5/2},$$

and $\tau_0(B_n)$ lies between the same bounds. The two outer terms are $n^3/54 + O(n^{5/2})$. $\square$

5. A numerical check

The script in the repository listed under Data availability computes three things. First, it computes Kemeny's constant of small barbells exactly, as $\operatorname{tr} Z - 1$ in rational arithmetic, and compares it with (2.2) and (2.3) evaluated in floating point; for example $\tau_0(B(3,2)) = 293/24$ and $\tau_0(B(6,7)) = 31363/222$. Second, for the barbells B_n it compares the bound of Proposition 4.1 with τ_2 and τ_0 computed from the spectrum. Table 1 shows the ratios to n^3, which approach $1/54 \approx 0.018519$ from above. Third, on 400 random connected graphs with 8 to 46 vertices (Erdős–Rényi graphs, random trees, chains of cliques and lollipops with extra edges) and the thresholds $D = 2, 3, \sqrt{n}, n/4, n/2$, it checks every inequality of Section 3: Lemma 3.1, the bound $B < 2nD$, Lemma 3.3 for each edge of a breadth-first spanning tree of the quotient, Lemma 3.2, the bound $R_{uv} \le \operatorname{dist}_T(q(u), q(v)) + \delta$, and (1.1). All of them hold.

n	a	ℓ	Proposition 4.1$/n^3$	$\tau_2(B_n)/n^3$	$\tau_0(B_n)/n^3$
30	10	11	0.019093	0.019387	0.021426
90	30	31	0.018718	0.018757	0.019266
300	100	101	0.018580	0.018583	0.018716
600	200	201	0.018549	0.018550	0.018614

Table 1. Balanced barbells B_n.

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This work was done by multiple AI agents, Claude (Anthropic) and Codex (OpenAI), connected to Math Brain [8] and working together inside the Brain framework [10], a shared, persistent memory in which the agents record claims, proofs, failed attempts and methods, retrieve one another's results, coordinate who works on what, and audit each other's work. For more on the details behind the Brain and Math Brain, see the Math Brain paper [9], the Math Brain repository [8] and the Brain framework repository [10].

Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, an ongoing project on the use of AI systems for research mathematics. In September and October 2026 a GPT-family model (OpenAI), running in the Codex command-line tool, found the proof of Theorem 1.1; other model instances checked it. In November 2026 Claude Opus 5.5 (Anthropic), working in Claude Code, checked the proof, wrote and ran the computations of Section 5, and wrote the paper. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

Data availability

No data were used. The code for the computations of Section 5, with its output, is available at `https://github.com/yspennstate/largest-kemeny-constant`.

The paper is part of the Math Brain project, whose public repository is `https://github.com/yspennstate/math-brain-public` [8].

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